

Polynomial Regression: BT2024167

VAR1 and VAR2 | Nested cross-validation model selection

1. Problems and Data

VAR1 predicts the Power Plant Steam Turbine Net Power Score from six operational inputs (x1-x6). VAR2 predicts the Subterranean Thermal Reservoir Thermal Anomaly Score from x1 East-West offset, x2 North-South offset, and x3 vertical depth offset. Both targets are y.

Variant	Train n	Test n	Inputs	Train y range	Duplicate rows train/test
VAR1	1000	1000	6	-7.873 to 11.900	0 / 8
VAR2	1000	1000	3	-29.609 to 39.518	0 / 10

Both datasets have numeric inputs and target, no missing values, and no ID/index columns. Feature columns are in the personalized CSVs; every train/test feature schema matched. Test labels were never read or used.

2. Method

Models use total-degree polynomial expansion (including interactions), followed by OLS, Ridge, Lasso, or ElasticNet. For regularized models, both raw polynomial features and StandardScaler-preprocessed features were compared. The outer evaluation uses five shuffled folds (random_state=7); within each outer training fold, a three-fold shuffled inner CV selects alpha (and ElasticNet l1_ratio). Thus each outer score is evaluated on rows not used to tune that fold's regularization. Selection minimizes outer-fold MSE; candidates within one standard error of the minimum are considered, then the lower-degree/sparser candidate is preferred.

Degree ranges: VAR1 2-7; VAR2 6-14. Lasso and ElasticNet use coordinate descent with warm starts and tolerance 1e-3. The final regularization strength is tuned using training data only, then each chosen pipeline is refit on all 1,000 training rows.

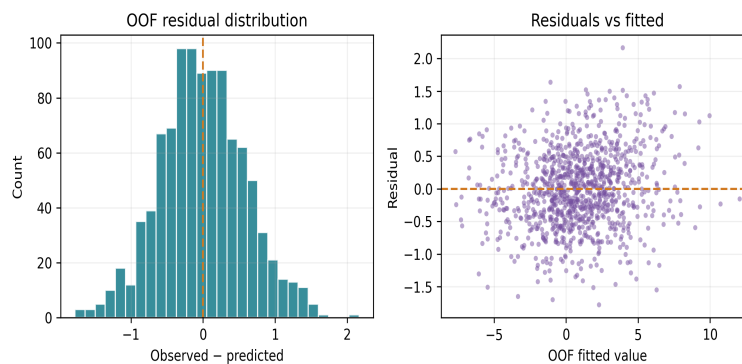
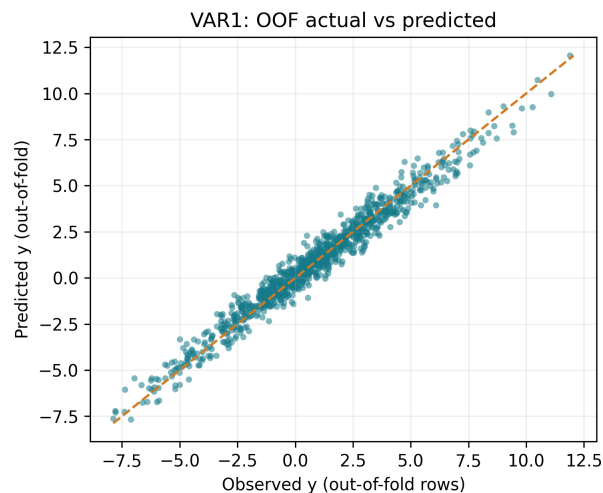
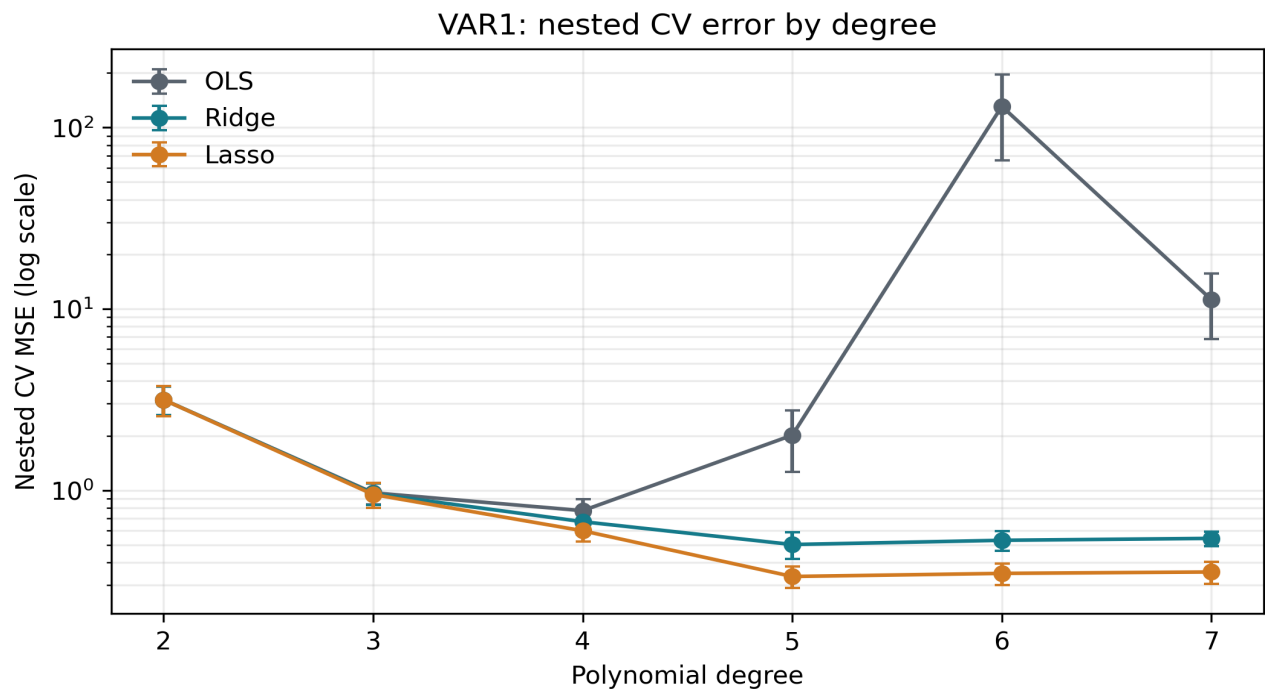
3. Selected configurations and original OLS baseline

Variant	Degree	Model / scaling	Alpha	Nested CV MSE mean ± SD	Nested CV R ²	OLS baseline MSE / R ²
VAR1	5	Lasso + StandardScaler	0.006	0.3491 ± 0.0384	0.9631	0.7717 / 0.9185
VAR2	10	Ridge	0.1	0.2329 ± 0.0134	0.9943	0.2556 / 0.9937

CV values are outer-fold estimates, not hidden-test scores. Relative to the original-degree OLS baselines (VAR1 degree 4, VAR2 degree 8), nested-CV MSE decreased by 54.8% for VAR1 and 8.9% for VAR2.

VAR1 — Degree Selection and Diagnostics

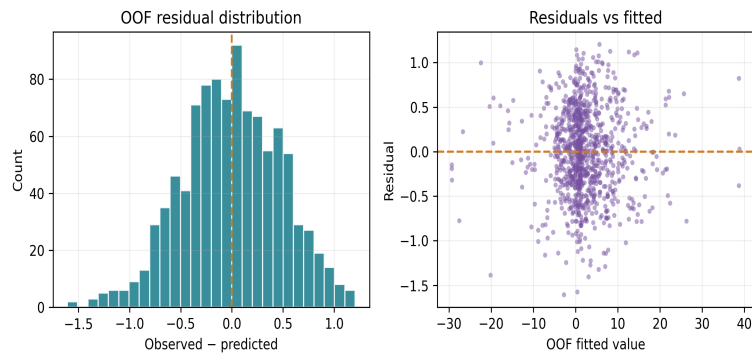
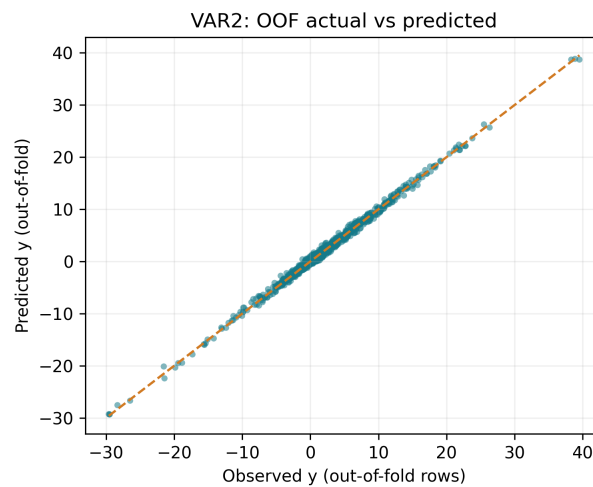
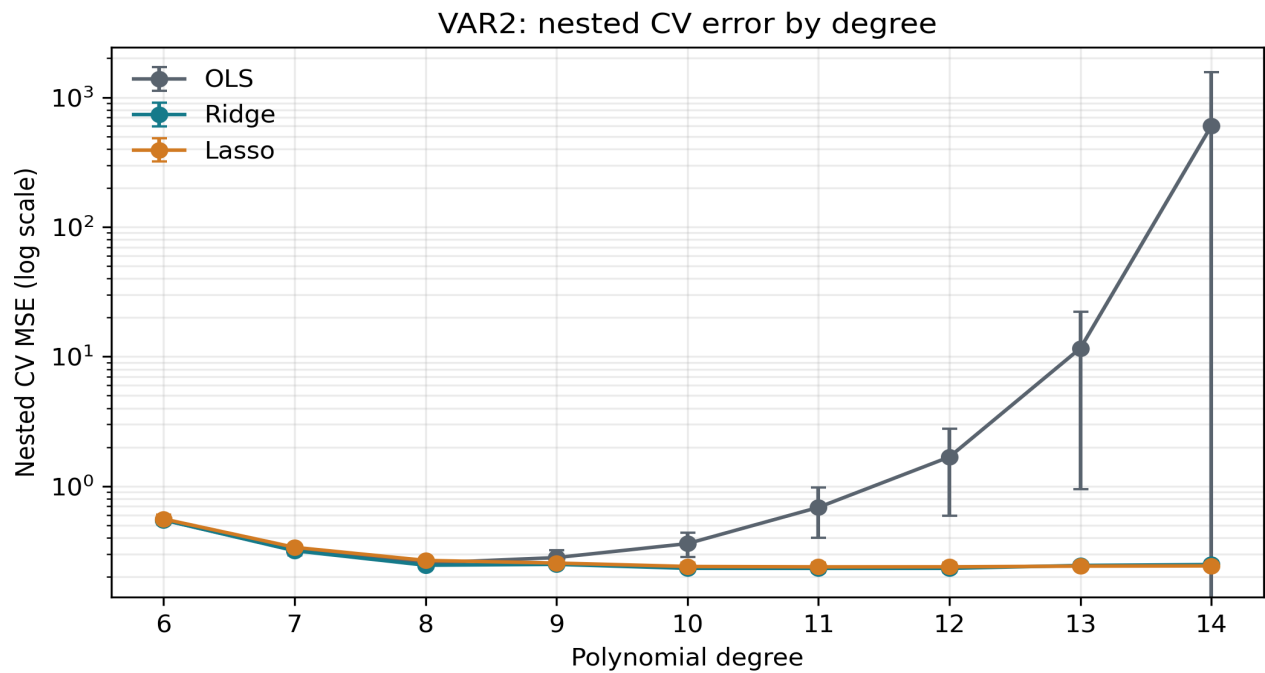
4. VAR1 results



The degree plot compares OLS against the best-scaled/unscaled Ridge and Lasso result at each degree; error bars show outer-fold standard deviation. The actual-predicted and residual figures use out-of-fold predictions from the chosen configuration. VAR1 Lasso retained 101.6 nonzero polynomial terms on average across the outer folds. The final full-data VAR1 fit retained 150 terms. The result table in `results/var1_nested_cv_results.csv` contains all candidate scores and alpha summaries.

VAR2 — Degree Selection and Stability

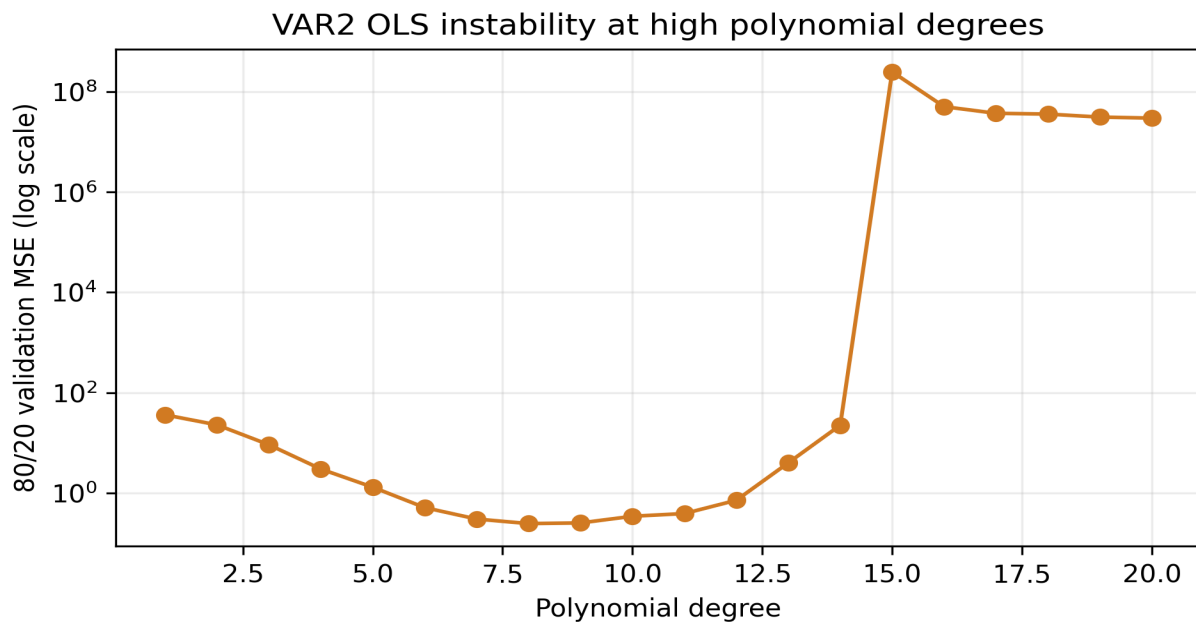
5. VAR2 results



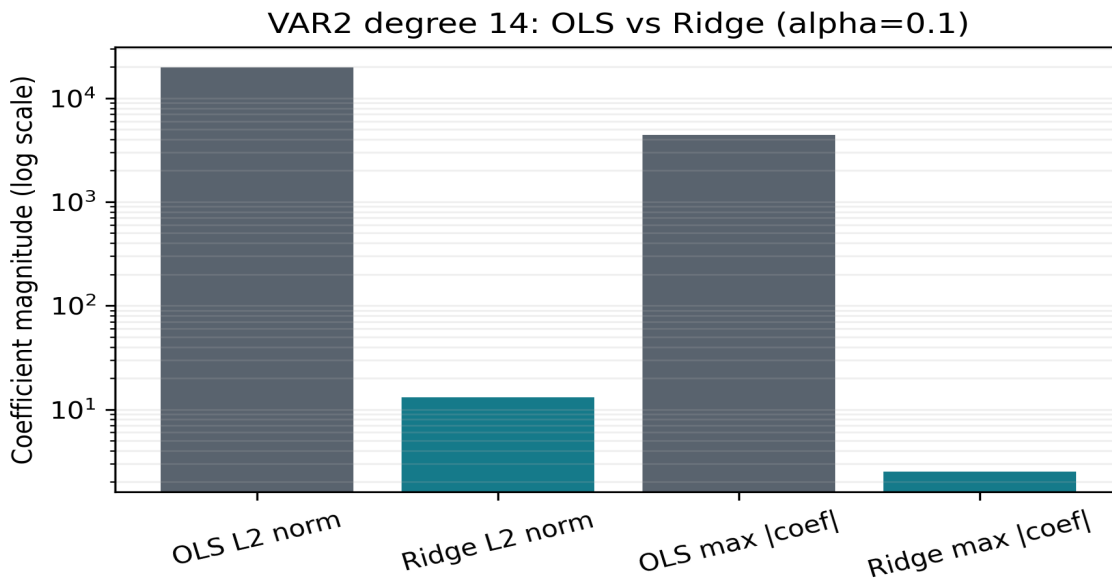
The degree plot compares OLS with the best-scaled/unscaled Ridge and Lasso at each degree. The remaining VAR2 numerical stability checks are shown on the next page; all validation diagnostics use training data only.

VAR2 — High-Degree Stability Checks

OLS becomes unstable at high polynomial degrees



The instability plot uses a separate 80/20 split of the training data (`random_state=42`). At degree 15, the design matrix has rank 793 of 815 columns and the smallest-to-largest singular-value ratio is $6.8\text{e-}17$. At degree 20, it has rank 793 of 1,770 columns, the ratio is $7.6\text{e-}17$, and validation MSE is about 29.9 million. These diagnostics illustrate why the maximum permitted degree is not a suitable default.



At degree 14, the chart compares coefficient L2 norm and maximum absolute coefficient for OLS and Ridge; Ridge alpha was selected using training-only five-fold CV. The numeric diagnostics are in `results/var2_ols_instability.json`.

Final Fit, Comparison, and Reproduction

6. Final model and predictions

After nested-CV selection, the chosen model was refit on all 1,000 training samples for its variant and used to predict all 1,000 matching test rows in the original order. The submission files contain only the numeric y column, with no index. Prediction verification and comparison with the archived OLS predictions are recorded in results/prediction_comparison.json.

7. Interpretation

VAR1 selected scaled degree-5 Lasso (alpha 0.006): its nested-CV MSE was 0.3491 versus 0.7717 for degree-4 OLS, and the final fit retained 150 nonzero polynomial coefficients. VAR2 selected degree-10 Ridge (alpha 0.1): its MSE was 0.2329 versus 0.2556 for degree-8 OLS. Degree 11 Ridge had slightly lower mean CV MSE (0.2327), but degree 10 was within one standard error and uses a smaller polynomial basis. These are CV estimates; no hidden-test performance is claimed.

8. Reproduction

Install the pinned packages from requirements.txt, then run from the repository root:

```
python train_final.py
python inference.py --variant var1
python inference.py --variant var2
```

The search writes nested-CV tables and configuration to results/, figures at 300 DPI to figures/, predictions to the repository root, and this report as BT2024167_Report.pdf. Inference uses the hardcoded selected configurations and does not rerun the search.